Sustainability standards and guidelines requirements for integrated management

Conference Paper · July 2013

6 authors, including:

Leandro Navarro Hundzinski

Edson Pinheiro de Lima
Federal University of Technology of Paraná

Sergio E. Gouvea da Costa
Federal University of Technology of Paraná

Carla Gonçalves Machado
Jönköping International Business School

SUSTAINABILITY STANDARDS AND GUIDELINES REQUIREMENTS FOR INTEGRATED MANAGEMENT

Abstract ID: OSPF10115

Abstract

Organizations are influenced by external demands, which can be represented by changes on the markets they are part of. Companies are operating on a new context, where the importance of sustainability has been highlighted by events like environmental problems, social responsibility issues and quality assurance of products and services. Those aspects are represented by the areas of the Triple Bottom-Line (3BL), which treat organizations operational performance on economic, environmental and social aspects. However, organizations are still trying to develop new performance measurement systems, which relate to the 3BL and that can be used for sustainable performance assessment and benchmarking. In order to address this need, this research will aim to develop a process that enables standards and guidelines to be used as tools for the establishment of a performance measurement system which can address the needs of integrated management on behalf of sustainability aspects. The method used on the development of this article utilizes concepts of the Process Approach methodology from Cambridge University, as well as Business Process Management Notation to represent processes for assessing performance. Those processes were built with a high emphasis on Performance Assessment frameworks and Key Performance Indicators (KPIs) theories. The processes generated have been applied in a project of innovation in sustainable management, in which its application enabled to work with requirements from various standards and guidelines in a practical context, through the usage of indicators and performance measurement. Exemplifying the application of this process, the article will aim to clarify the usage of indicators based on standards and guidelines as a tool for sustainability assessment, through an integrated management of an organization set of standards and guidelines.

Keywords:
Sustainable operations, Standards, Process approach, Sustainable Performance Measurement System.

1 INTRODUCTION

Organizations find themselves in an ambient of growing competition and exigency. In this context, certifications and standardizations represent a validation, given by external institutions, that the operations associated with the company are in agreement with widely accepted criteria from its clients and suppliers about quality, reliability, environmental and social responsibility [1][2].

However, the management of requirements for certain standards and guidelines needs to be structured, due to the fact that those does not represent formalized indicators, which establish metrics capable of being measured, monitored and managed. Neely et al. [3] brings focus on the fact that defining a performance measure involves more than simply identifying formulas. Requirements need to be aligned to the organizational structure before being able of measuring performance, as it may lack definitions of measure, purpose, actors, source of data, target and frequency [3].

Parmenter [4] identify KPIs as a set of measures that focuses on the most critical aspects of organizational performance for the current and future success of the organization. As such, defining right KPIs is essential for organizations to structure its performance management system in relation to its most important strategies and business orientation.

Hubbard [5] emphasizes the multi-faceted nature of organization performance and the increasing complexity in measuring performance due to changes on stakeholder expectations about companies' economic, social and environmental responsibilities. Fiksel et al. [6] also detects this complexity by highlighting the difficulties faced by companies to track their progress towards sustainability and to communicate that progress to stakeholders.

For Schwarz et al. [7], it is necessary that 'practical, cost-effective ways to assess performance and measure progress' be developed in order to integrate sustainability into organizations operations.

Those authors highlight the crescent need of business to understand how to implement the concepts of sustainable development [7], represented by the development of models and frameworks for enabling measurement towards sustainability.

This aim of this study is to address those challenges faced by organizations when assessing sustainable performance, by reducing the amount of effort necessary for an organization to manage its sustainable performance system, by taking into consideration already considered aspects of standards and guidelines into the organization.

The objective of this paper is to establish a procedural model for the conversion of requirements from standards and guidelines into indicators, based on the business process modeling (BPM), in order to assist organizations in aligning its performance management system with its strategies and business objectives, translated into sets of standards and guidelines.

The article is constructed based on the best practices of performance measurement [3][8][9], sustainable performance measurement [6][10], BPM orientation [11] and standards and guidelines management. The BPM Suite (BPMS) used for the generation of processes models was BonitaSoft solution, Bonita Open Solution, an Open Source tool for the modeling of business process [12].

As a consequence, a simplification of processes is expected to be reached, by integrating standards and guidelines management into the development of a functional sustainable performance management system, with relevant KPIs and the ability to contribute to the continuous improvement of organizations.

2 METHODOLOGY

The development of this article was based on the Process Approach from University of Cambridge, which aims to operationalize conceptual models into usable applications [8]. Neely et al. [8], through the application of surveys, determined that organizations which employ formal processes in order to design its performance measurement system facilitate the following tasks:

- Decide what they should be measuring
- Decide how they are going to measure it
- Collect the appropriate data
- Eliminate conflict in their measurement system

Due to those facilitations, the BPM methodology was then applied in order to represent and operationalize the model obtained by the Process Approach. According to Leite and Rezende [13], BPM is an evolution of the simple workflow, because it does not only allows the graphical representation of processes and automate workflow, but it also integrate processes by sending information through automated activities or human activities, capturing results in order to define pathways and monitor the executed process. That makes possible for organizations to assess its performance and mitigate risks from its operation in a very dynamic way.

The strategy used for the conduction of this research was to base its contents on profound studies on performance measurement systems and the needs organizations have in relation to those systems. Based on studies that are relevant to the context of performance measurement and sustainable performance measurement systems, the strategy aims to ensure that the output of this research optimizes organization performance assessment related to the areas of the 3BL, by allowing a clearer understanding of the company needs in relation to standards and guidelines management and sustainable performance assessment metrics.

The planning for this research is related to its application in a project of innovation in sustainable management. Its application is intended to facilitate the work with requirements from various standards and guidelines in a practical context, through the generation of indicators and performance measurement models. The development of the research is structured by action research, which can be considered a variation of case studies [14], and is defined by Peter Reason and Hilary Bradbury [15] as:

'A participatory, democratic process concerned with developing practical knowing in the pursuit of worthwhile human purposes, grounded in a participatory worldview which we believe is emerging at this historical moment. It seeks to bring together action and reflection, theory and practice, in participation with others, in the pursuit of practical solutions to issues of pressing concern to people, and more generally the flourishing of individual persons and their communities.'

The outputs of the research will then be detailed by its application and the results obtained from it.

3 SUSTAINABILITY AND MANAGEMENT MODELS

The management of requirements from several standards and guidelines in a sustainability oriented organization increased the demand for management models to be created. Parmenter [4] emphasizes the fact that many companies are referring to their measures as KPIs, but that very few of those organizations monitor their true KPIs. Neely et al. [3] as well highlights the importance of performance measurement because of its significant contribution to the implementation of strategies by managers. Accordingly to the Neely et al., performance measures should be used to, as suggested by recent developments:

- Clarify strategy.
- Communicate and drive strategy.
- Check implementation of strategy.
- Challenge strategy.

Companies are constantly increasing their sustainability awareness, which implies changes in corporations' structure. To Labuschagne et al. [16], "Business sustainability entails the incorporation of the objectives of sustainable development, namely social equity, economic efficiency and environmental performance, into a company's operational practices."

Searcy [17] indicates that integration into operational practices is a substantial challenge in applying sustainable development at the corporate level. The author strengthens Parmenter's view [4], by addressing that 'In some jurisdictions, corporations may be required to develop sustainability or related reports[…] However, the fact that a corporation develops a report does not necessarily mean that senior managers use sustainable development indicators as a key input in their decision-making processes.'

It can be conclude that companies are being challenged to improve their identification and development of KPIs, in order to establish consistent changes into the direction of more sustainable operations.

3.1 Standards and guidelines for sustainable development

The ISO's (International Organization for Standardization) portfolio surpasses 19100 standards, which provides solutions in environmental, economic and social dimensions. [18]

Public, governmental and commercial organizations have become more aware of sustainability in the last five years. ISO confirms that by highlighting that [1]:

'As a result, an explicit commitment to sustainability has been reflected in standards for those areas of economic activity where it is most pressing, notably construction and the management of water, waste and energy. For the near future it is likely that standards addressing supply chain issues, employment and, in the widest sense, management systems, will become established as popular and effective tools […] In the longer term, we can expect sustainability to become a fundamental principle for ISO standards in just the same way as market relevance.'

Even though ISO's influence on standards is of major importance, sustainability oriented standards are not represented only by ISO, but by a lot more of organizations. Accordingly to the Sustainability Compendium [19], standards "foster a deep reflection regarding management tools to be used in order to ensure sustainable evolution planning." Other important standards referred by the Sustainability Compendium are:

- SA 8000 (social rights)
- OHSAS 18001 (risks/accidents)
- AA 1000 (accountability)
- CE EMAS (environment)
- BS 8800 (decent work conditions)
- BS 8855 (environment)

Not only standards guide a company in direction to sustainability, but there are also a lot of different guidelines in a huge diversity of documents, intended to give organizations orientation in the aspects of sustainable development. Those guidelines usually have the objective of orienting the organization through a certain objective of sustainability. Organizations trying to integrate sustainability into their operations are always in search for best guidelines and practices, as those represent an external validation of their processes to its stakeholders.

One of those guidelines is the United Nations Global Compact, which asks companies to embrace, support and enact, within their sphere of influence, a set of core values on human rights, labour standards, environment and anti-corruption. [20]

Naveh and Marcus [21], in their article, focus on the effects on measures of business and operational performance, due to the implementation of ISO 9000. According to the conclusions reached by Naveh and Marcus [21], the advantages of implementing a managerial practice such as ISO 9000 depend on how the practice is implemented. To the authors, "competitive advantage in the operations area can be achieved if the standard is installed—externally coordinated with suppliers and customers and integrated with practices the company already has in place—and used—both in daily practice and as a catalyst for change."

We can then conclude that standards and guidelines can be used as triggers for optimization of performance measurement on companies, as long as it supports the business strategies. This is due to the fact that standards and guidelines are able to:

- Assess the organization's Value Chain;
- Reflect business and strategic objectives of organizations;
- Relate to an ever-changing environment, with new internal and external demands;

Business sustainability must be practical but flexible. It should be able to adequate with constantly changing business needs, costumers expectations and needs of society [22]. This statement bring us the necessity of addressing performance measurement as a constantly updatable activity.

4 SUSTAINABILITY PERFORMANCE MEASUREMENT

Neely et al. [3] identify performance measurement as the processes used to quantify efficiency and effectiveness of purposeful action. Those processes, according to Baker [9] can be seen as a series of steps that can be separated into three phases: Plan; Implement; Review and Act. This three-phase structure is analogous to ISO 14031 framework for Environmental Performance Evaluation [6], in which the PDCA (Plan-Do-Check-Act) structure is embedded [23].

Fiksel et al. [6] proposed a Sustainability Performance Measurement Process, based on those three phases. The authors aim to design a performance indicator framework with well-defined actors for the measures assessment, which is constantly updated to reflect changes on knowledge and on the state of art.

Neely et al. [3] developed a method for business assessment that also includes stages of planning, doing, checking and acting. The authors objective are to establish a balanced set of performance measures that is consistent with business objectives and that can be analyzed in order to identify how business performance can be improved.

We can sum those viewpoints into the following image, which is an overview of the stages from the processes proposed by the authors:

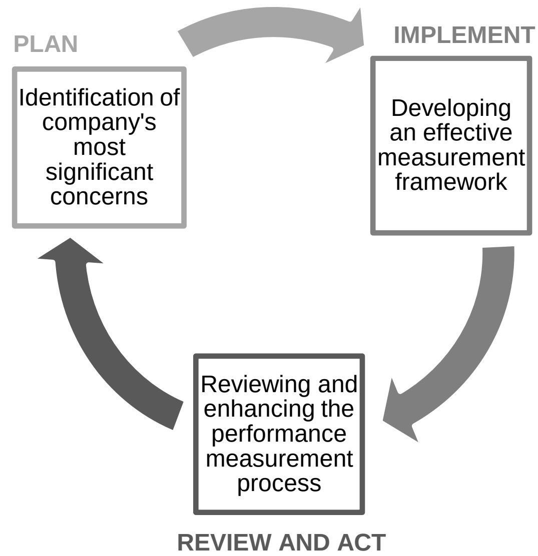

Figure 1: Overview of a Sustainability Performance Measurement Process [3], [6]

Those models highlight the importance of implementing a closed-loop system, with the possibility of obtaining feedback after some period of application into the company. That is a consistent aspect of those systems, due to the difficulty of assessing sustainability from the organizations processes because its concepts are still evolving and changing [6].

On the description of sustainable development oriented assessments, Hacking and Guthrie [10] affirm that those assessments are invariably addressed to adopt more ambitious goals and to provide more strategic assessments. Those goals, according to the authors, should also contribute to sustainable development by proactively enhancing positive impacts, instead of only avoiding negative impacts on organizations.

4.1 Integrating standards and guidelines into the organization sustainable performance measurement

Sustainability concepts are in constant changes, triggered by the increased awareness from stakeholders in relation to sustainability and the creation of gradually more consist studies about the subject.

This movement is similar to the development of new standards and guidelines, which mirrors on the emerging needs of organizations, stakeholders and markets. Some examples are ISO 26000 and ISO 50001. Those stantards represent the recent need for the establishment of a structure in respect to social responsibility problems [24] and energy distribution[25], which represent bottlenecks on the pathway to sustainability.

In order to identify those elements in standards and guidelines, the model from Figure 1 can be oriented to standards and guidelines and its application in relation to the 3BL:

Table 1: Standards and guidelines applied to Performance Measurement processes

Plan
Standards and guidelines utilized by the organization are used to identify company's most significant concerns, in respect to its needs. Their range is very diversified, and can orient activities in all areas of the Triple Bottom Line. Choosing the right set of requirements can help organizations to:
•*Establishing clear business objectives*
•*Work with customers need*
•*Assists to constantly adapt strategic positioning*
•*Orient the organization in relation to a environmental management system*
•*Include social responsability into the organization performance objectives*
•*Stimulate shared value initiatives*
•*Driving performance in direction to strategic objectives*
•*Developing measures to drive business performance*
Implement
The chosen set of requirements must be treated in respect to its needs of compliance, represented by indicators which can assess performance or control achievement of specific objectives. Some example of objectives that those indicators can help a company achieve are:
•*Generate less environmental waste*
•*Increase productivity*
•*Creation of social responsible outcomes*
•*Increase in workers satisfaction*
•*Raise quality standards*
•*Aim for cleaner production processes*
Review and Act
Internal and external auditing processes, certification renewal and new procedures for the maintenance of standards and guidelines relate to reviewing strategic orientation and verifying implementation effectiveness. Those aspects relate to review intervals of standards and guidelines and can orient a new improvement cycle in respect to:
•*Right choice of measures*
•*Determining new business objectives*
•*Optimizing processes*
•Reviewing key performance indicators

However, the management of requirements from certain standards and guidelines needs to be structured, due to the fact that their requirements may not represent indicators, which can be measured, monitored and managed. According to Kennerley e Neely [26], a performance indicator:

- Must be derived from the strategy (can be understood as the need for action due to a specific requirement);
- Be easy to understand (of easy use for managerial activities);
- Provide accurate and regular feedback;
- Be based on quantitative data that can be influenced or controlled by the person or group responsible for its performance;
- Be objective (not based on opinion, so that its elements are able to be proven)
- Be accurate (its capacity to relate and translate the measures that are needed to be measured)
- Be related to specific targets: measures to be achieved, compliance with standards, monitoring and supervision.

The following list contains examples of strategic objectives that can be translated into performance indicators:

- Is the scope of the environmental management system properly defined and documented? (From ISO 14001, clause 4.1 - Values: yes or no) [27]
- Does the organization respect human rights and recognizes both their importance and their universality? (From ISO 26000, clause 4.8 – Values: yes or no) [24]
- Is there evidence that the company invests in social programs outside of the organization? (Values: yes or no)

In order for a set of requirements to be used as indicators, they need to be converted. The flow of the conversion process is integrated with the steps of identification of demands for each requirement, assignment of ways for measuring accordance and validation of the indicator generated by the process.

The technique involved in the method of modeling is based on studies from the University of Cambridge, through the methodology of Process Approach, which seeks to operationalize conceptual models in usual applications.

The following model represents that process, and was constructed using the Business Process Management Notation (BPMN), with the assistance of BonitaSoft BPMS [12]:

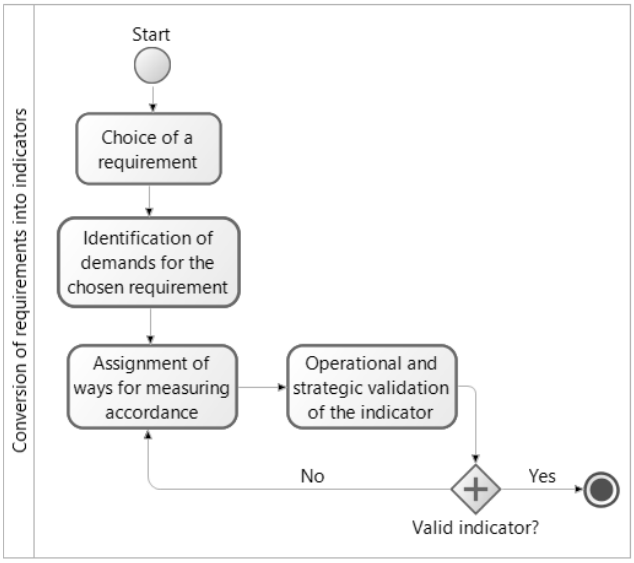

Figure 2: A process for the conversion of standards and guidelines requirements into indicators

The process on Figure 2 allows the development of a set of indicators focusing on the content of various requirements from standards and guidelines. Its objective is to establish a process for the adequacy of those requirements into the measurement of organizational performance, in relation to sustainable aspects.

4.2 Application of the process for the conversion of standards and guidelines into indicators

The process for the conversion of requirements into indicators will be covered from subchapter 4.2.1 to subchapter 4.2.4, exemplifying the basis for those processes to be followed.

4.2.1 Step: Choice of a requirement

The person responsible for this process must understand the reasons behind the implementation of the set of standards and guidelines the organization works with. This is necessary so that the KPIs can be identified within that set requirements. The person must address the companies KPIs by choosing from those requirements, being that in some cases more than one requirement may be necessary in order to attend the needs of an indicator, and vice-versa.

This step involves reading and profound understanding of how the analyzed requirement demands relate to the organization, in order for measures of efficiency and effectiveness to be attributed.

The Balanced Scorecard (BSC) is a strategy performance management tool that has the objective of aligning business activities with the vision and strategy of the organization, improve communication with stakeholders and utilize strategic goals in order to monitor organizational performance [28]. As such, it can be used in this process to help the KPIs related to requirements to be identified.

Parmenter [4], based on recent studies of Kaplan and Norton about the BSC, establishes a six-perspective BSC, instead of the original four perspective view. The importance of employee satisfaction and environmental/community perspectives are considered on this version, which incorporates all 3BL issues on the BSC:

Figure 3: Six-Perspective Balanced Scorecard [4]

This tool can be used in order to clearly identify the company strategies and business orientation, facilitating the choice of significant requirements to be used on the Sustainable Performance Measurement System. When analyzing a specific standard or guideline, the BSC can provide a clear picture on how that standard or guidelines relate with the organization, facilitating the task of identifying significant requirements to be integrated into the organization management.

4.2.2 Step: Identification of demands for the chosen requirement

Requirements have demands that must be addressed in order for it to be in accordance with the needs of standards and guidelines. In order to guarantee accordance, those demands have to be clearly identified, in such way that metrics and indicators have a direct relation to its needs.

For that step to be successful, the systems implemented by standards and guidelines must be very clear on its reasons for implementation and areas involved. That will allow for the next step to be consistently built.

The person responsible for this step must clearly identify how the chosen requirement(s) relate to the organization structure and how it is managed in relation to standards and guidelines applied on the company. It must be clarified who is involved with this requirement, which areas of the organization control it conformity, which impacts does it bring to the organization and why is it important to the business performance [3].

It will then be clear on what area of the 3BL that requirement will be situated with. This step is then concluded. The next step will define where metrics and controlling will be added to the requirement structure in order to establish a valid indicator, capable of assessing performance.

4.2.3 Step: Assignment of ways for measuring accordance

Indicators can be used for controlling accordance of requirements and for assessing performance, by the evaluation of its data. Depending on the content of the requirement, an indicator may not be able to have complex metrics, but simply relate to the state of compliance to that requirement.

The requirement content in relation to the standard or guideline it is from will determine whether it will be able to be assigned with simple logical value for conformance control and verification (Yes or No / Have or Do not Have / etc) or dedicated metrics capable of addressing specific performance objectives, not only accordance with standards and guidelines.

An execution of this step can be verified on ABNT NBR ISO 14031:2004 [23], which details how environmental performance assessment for a company can be addressed. According to that document, data from indicators can be:

- Measurement or direct calculation: basic data or information. E.g.: tons of contaminants emitted.
- Measurements or relative calculation: data or information compare or related to another parameter. E.g.: tons of contaminants emitted per year.
- Indexed: data or descriptive information converted to units, or to a way that relates information to a pattern or reference chosen. E.g.: tons of contaminants emitted per year as a percentage of the total emissions.

- Aggregate: descriptive data or information of the same kind, but from different sources, collected and expressed as a combined value. E.g.: tons of contaminants emitted per year from the production of a product, determined by the sum of emissions from all the installations that produce that product.
- Weighted: descriptive data or information modified by the application of a factor related to its significance.

ABNT NBR ISO 14031 [23] categorize indicators as indicators of management performance (gives information about management efforts to influence operations performance) and indicators of operational performance (gives information about operations performance). Both kinds of indicators can be related to the kinds of data related above.

That classification in both aspects (data and function) will be used in order to identify the indicator's function on the sustainable performance measurement system, in order to establish a consistent view on indicators.

4.2.4 Step: Operational and strategic validation of the indicator

After the indicator has been defined, it needs to be validated within the context of the organization. The stage of operational and strategic validation of the indicator, based on studies conducted by Andy Neely et al. [3], the following checklist consists in checking if the indicators are:

- Consistent: Meets demands of the requirement, as to ensure its compliance.
- Exact: Allow a logical value to be assigned by analyzing the indicator, which results in compliance or non-compliance with the requirement;
- Updatable: The data provided to an indicator must be updatable, periodically or when changes occur in the organization. This may mean that an indicator which was previously in accordance to its requirements, can no longer be in accordance, and vice versa;
- Justifiable: The indicator must allow the causes of compliance or non-compliance with its requirement to be identified, in order to enable measures and proceedings to be taken.

This step allows assessing if the analyzed indicator meets the operational and strategic needs for which it was proposed, thus validating its managerial functionality.

4.2.5 Maintaining the process consistency

The steps of the process from Figure 2 must then be applied to all requirements to which the organization wants to integrate its performance management system.

When converting requirements of different standards and guidelines focused on sustainability, the organization can cover the areas of the 3BL, covering the organization's value chain and facilitating management to become an integrated and sustainability focused activity, through relevant indicators and metrics.

An organization that follows those steps can then simplify the management of various market demands brought by sustainability, by translating its needs into standards and guidelines that can then be used to create an integrated sustainable performance measurement system.

Working with indicators allows transforming information resulting from the implementation of standards and guidelines into quantifiable data, allowing managers to discover which requirements are not being met or did not achieve satisfactory performance.

Therefore, it represents an effective tool for structuring an integrated management for sustainability oriented by standards and guidelines requirements and changes on stakeholder views.

Mosse and Sontheimer [29] study depicts a logical framework methodology for helping stakeholders and project designers, among project specific objectives, to establish proper objectives, define indicators of success and identify means for project accomplishments verification. In one application of the logical framework, Mosse and Sontheimer [29] utilize four aspects for the verification of a project specific context:

- Narrative summary impact: what objective are you trying to achieve?
- Performance indicators: what is the quantifiable data that represent that objective?
- Monitoring and supervision: where progress towards the defined objective can be verified?
- Assumptions and risks: what is necessary for this objective to be implemented and what risks implementation may undergo?

This framework can be used to give more guidance on the process of elaboration of a sustainable performance management system, controlling the implementation and clarifying the results which are desired to be obtained after that integration.

4.3 Applications and Results

This research was applied to enable the work with a set of 6 standards and 4 guidelines related to sustainable development:

- Standards: SA 8000, ISO 14001, OHSAS 18001, ISO 26000, ISO 9001 and ISO 50001.
- Guidelines: Global Compact, Agenda 21 Brazil, United Nations Millennium Development Goals, AA 1000 Accountability.

The application followed the steps detailed on chapters 4.2.1 to 4.2.4. ISO 14001 will be considered as a case for its application.

The first step, detailed on 4.4.1, allowed understanding the reasons behind the implementation of ISO 14001. The six-perspective Balance Scorecard from Figure 3 has the objective of helping managers clarify the needs of the analyzed standard in relation to sustainability:

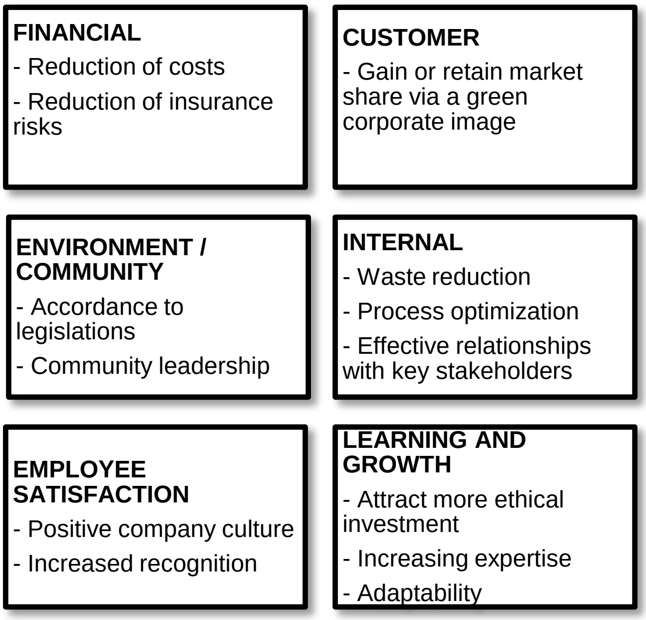

Figure 4: Six-Perspective Balanced Scorecard applied to ISO 14001– based on [4] and [30]

Iteratively, each requirement of the ISO 14001 will need to be addressed in order to maintain the system's conformity. This example will go through the Clause 4.4.2 (refers to Competence, training and awareness), which must ensure that collaborators who engage into activities that may cause significant environmental impact must be competent and receive training [27]. To correctly address this requirement, the organization needs to [30]:

- Ensure that training needs are identified;
- Ensure that these planned needs are met;
- Verify that the training has achieved its purpose – increased awareness;
- Verify that following training, the individual is competent at applying the awareness gained to their particular job.

This particular requirement can be related to the Employee satisfaction (positive company culture) and Learning and Growth (increasing expertise) aspects of the six-perspective Balance Scorecard applied for ISO 14001, on Figure 4. It can also be related to the 3BL Sustainability aspect, due to its capacity of skill enhancement with an environmental orientation to its objectives.

At next step execution, identification of demands for the chosen requirement, the manager must understand the organization operations, in order to be able to work with requirement demands. Managers need to be engaged on this activity, gathering information in order to identify actors, which areas are involved and how its effects on performance is assessed.

The following tables, detailed in Neely et al. [3], exemplify the application of the step from chapter 4.2.2. The Table 2 defines objective 1 and 2, and Table 3 defines responsibilities in relation to those objectives:

Table 2: Clause 4.4.2 objectives [3]

Objectives				
			Target	
	Description	Priority	Improvement	By when?
1	Identify and meet training needs	50	% more investment in training	Per semester
2	Increase competence by training	50	% increase in enviromental impact awareness	Next certification renewal cycle

Table 3: Responsible staff for Clause 4.4.2 objectives [3]

	Responsibilities and contributions						Check / develop measure
	Mnf.	Sales	Dev.	Fin.	H.R.	Qual.	
1					100%		H.R. employee
2	20%		20%		20%	40%	Quality Assurance area

Measures must then be developed by associating requirements with aspects of classification of its data listed on chapter 4.2.3 (measurement or direct calculation, measurements or relative calculation, indexed, aggregate or weighted). The classification of the data will assist in attributing the requirement with an indicator structure, associating it as an indicator of operational performance or of management performance.

The following table aligns the objectives from Table 2 to possible measurements:

Table 4: Indicator structure for Clause 4.4.2 [23]

	Measure	Data classification	Indicator category
1	Sum of Investments on trainings about environmental aspects of operation offered per collaborators per year	Indexed	Indicator of management performance
2	Number of events related to significant environmental impact operations per year	Measurements or relative calculation	Indicator of operational performance

The data obtained must enable performance to be measured and conformity aspects to be addressed. This verification is done on the next step, detailed on chapter 4.2.4.

The set of indicators obtained must be validated. This validation will be conducted by the application of the checklist derived from Neely et al. [3]:

- Is (are) the indicator(s) consistent? Yes.
- Is (are) the indicator(s) exact? Yes.
- Is (are) the indicator(s) updatable? Yes.
- Is (are) the indicator(s) justifiable? Yes.

The output of the application of the process from Figure 2 is a valid indicator for the assessment of the clause 4.2.2 from ISO 14001 and its requirements. The indicator can then be integrated into the performance assessment of the organization, allowing the integration between the clause and the performance measurement system. This process can then be reiterated, in order to fully integrate a considered standard or guideline.

The new indicator orientation to sustainability is also identified, as of the analysis provided by Figure 4. In this particular case, the indicator originated from Clause 4.4.2 (which refers to competence, training and awareness), can be related Employee satisfaction and Learning and Growth perspectives, as well as represents the Sustainability area of the 3BL, by unifying socio-economic (training), socio-environmental (environmental responsibility and regulation) and eco-efficiency (resource efficiency and environmental responsible processes development) areas.

The model developed is being applied in a project of innovation in sustainable management. Its application enabled to work with requirements from various standards and guidelines in a practical context, through the generation of indicators and performance measurement models. This research received deep contribution from the research conducted by Machado et al. [31], which established a process for the formulation of indicators for sustainable operations management. Both researches developments were applied to the same project, allowing the production of a tool for management support for sustainability assessment.

5 CONCLUSION

This research allowed a clearer understanding on the process of managing standards and guidelines by organizations.

It was successful in establishing a procedural model for the conversion of requirements from standards and guidelines into indicators, facilitating the creation of significant KPIs by organizations, in relation to its strategic orientation.

The model was applied in a project of innovation in sustainable management [31], allowing to iteratively work with a set of 6 standards and 4 guidelines, through the reiteration of the process for the conversion of standards and guidelines requirements into indicators. This assisted on the creation of a sustainable performance management system which assesses performance based on standards and guidelines and its relation to the areas of the 3BL.

Ultimately, this research enable to generate a simplification of the process of sustainability assessment, through the integration of standards and guidelines management into the development of a functional sustainable performance management system, by attributing relevant KPIs and following a continuous improvement system with phases of planning, implementation, reviewing and acting.

6 ACKNOWLEDGMENTS

Authors wish to thank Curitiba Agency through its ‘Curitiba ISS Technology Incentive Program’ that is regulated by Supplementary Law No. 39/01 and Municipality Decree law No. 646/2011, for financial supporting this research project (Protocol number 01-136095/2011). Also the authors wish to thank PUC Agency for Science, Technology and Innovation - PUCPR and Sofhar Gestão&Tecnologia for supporting this research.

7 REFERENCES

[1] Danvers M., 2012, Building blocks – ISO standards as powerful tools for taking action, International Organization for Standardization. Available at: http://www.iso.org/iso/home/news_index/news_archive/news.htm?Refid=Ref1522

[2] Rosing, M. v., Schell, H. v., 2010, Creating a better future with the right Business Sustainability Management Framework. Business Management Team.

[3] Neely A., Bourne M., Mills J., Platts K., Richards H., 2002, Getting the measure of your business. Cambridge: Cambridge University Press.

[4] Parmenter D., 2007, Key Performance Indicators. Indianapolis: John Wiley & Sons.

[5] Hubbard G., 2009, Measuring Organizational Performance: Beyond the Triple Bottom Line, Business Strategy and the Environment, 19, 177-191.

[6] Fiksel J., MacDaniel J., Mendenhall C., 1999, Measuring Progress Towards Sustainability Principles, Process, and Best Practices, Greening of Industry Network Conference - Best Practice Proceeding.

[7] Schwarz J., Beloff B., Beaver E., 2002, Use Sustainability Metrics to Guide Decision Making, ‖ Chemical Engineering Progress, 98, 58-63.

[8] Neely A., Mills J., Platts K., Gregory M., Richards H., 1996, Performance measurement system design: Should process based approaches be adopted?, Int. J. Production Economics, 1996, 46-47, 423-431.

[9] Baker, K., 1999, The Development of Organization and Program Performance Indicators. Pacific Northwest Laboratory.

[10] Hacking T., Guthrie P., A framework for clarifying the meaning of Triple Bottom-Line, Integrated, and Sustainability Assessment, Environmental Impact Assessment Review, 2008, 28, 73-89.

[11] Baldam, R., Silva, H. P. da, Valle, R., 2007, Gerenciamento de Processos de Negócios. BPM – Business Process Management. São Paulo: Érica.

[12] BonitaSoft S.A., 2012, Bonita Studio – Community Edition version 5.7.2. Available at : http://www.bonitasoft.com

[13] Leite L. de O., Rezende D. A., 2007, Gestão Corporativa por Processos na Administração Pública Municipal: Estudo de Caso da Implantação do Business Process Management (BPM) no Instituto Curitiba de Informática, EnADI, Florianópolis (SC).

[14] Westbrook R., 1995, Action research: a new paradigma for research in production and management, International Journal of Operations and Production Management, 15, 6-20.

[15] Reason P., Bradbury H., 2001, Handbook of action research: Participative inquiry and practice. London: Sage Publications.

[16] Labuschagne C., Brent A. C., Erck R. P. G., 2005, Assessing the sustainability performances of industry, Jornal of Cleaner Production, 13, 373-385.

[17] Searcy C., 2009, The role of Sustainable Development Indicators in Corporate Decision-making, IISD, 1-23.

[18] ISO, 2012, Rio+20 - Forging action from agreement, International Organization for Standardization.

[19] Sustainability Compendium, Standards and Certifications, Sustainability Compendium – Social and Environmental Responsability Management Tools. Available at : http://www.institutoatkwhh.org.br/compendio/pdf/novo/en/compendium_part_03.pdf

[20] The Global Compact, 2011, Corporate Sustainability in the World Economy, United Nations Global Compact.

[21] Naveh E., Marcus A., 2005, Achieving competitive advantage through implementing a replicable management standard : Installing and using ISO 9000, Journal of Operations Management (JOM), 24, 1-26.

[22] Developing Value – The business care for sustainability in emerging markets, SustainAbility, International Finance Corporation, Ethos Institute – Business and Social Reponsibility Available at: http://www.greenbiz.com/sites/default/files/document/O16F30165.pdf

[23] ISO, 2004, ABNT NBR ISO 14031: Gestão ambiental – avaliação de desempenho ambiental – Diretrizes. Rio de Janeiro: Associação Brasileira de Normas Técnicas.

[24] ISO, 2010, ISO 26000: Guidance on social responsability. International Organization for Standardization.

[25] ISO, 2011, ISO 50001: Energy management systems – Requirements with guidance for use. International Organization for Standardization.

[26] Kennerley M., Neely A., 2003, Measuring performance in a changing business environment, IJOPM, 23, 213-229.

[27] ISO, 2004, ISO 14001: Environmental management systems – Requirements with guidance for use. International Organization for Standardization.

[28] Balanced Scorecard Institute - How a Balanced Scorecard Can Help Your Organization. Available at: http://www.balancedscorecard.org/Portals/0/BalancedScorecardInstitute-BSC.pdf

[29] Mosse R., Sontheimer L. E, 1996, Performance Monitoring Indicators Handbook, World Bank Technical Paper n º334.

[30] Whitelaw K., 2004, ISO 14001 Environmental Systems Handbook, Second Edition. Elsevier Ltd..

[31] Machado, C. G., Pinheiro de Lima E., Gouvea da Costa S. E., Cestari J. M. A. P., Kluska R. A., Hundzinski L. N., 2012, Indicators formulation process for sustainable operations management, In: ICPR Americas. Santiago, Chile.